

What Determines the Inflation-Adjusted Value of NFL Contracts for Offensive Skill-Position
Players?

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I. Introduction

The National Football League is a labor market in which firms compete for talent under unusually tight constraints. In this market, the firms are the teams. When teams decide which contracts to offer, they evaluate players whose output is difficult to observe perfectly, commit to multi-year agreements without being able to forecast future performance and health with certainty, and operate under a hard salary cap. That institutional setting makes the NFL a useful applied case for economic analysis. Contracts help show how productivity is translated into compensation, how labor-market institutions shape pay, and how different forms of information affect the price of labor. These issues are especially important for offensive skill positions because quarterbacks, running backs, wide receivers, and tight ends face different market expectations, different degrees of replaceability, and different forms of demand.

This paper examines the relationship between a set of observable characteristics and the inflation-adjusted average annual value of NFL contracts for offensive skill-position players. More specifically, it tests whether contract-year efficiency and usage are associated with real contract value after controlling for other relevant factors. The dataset contains 372 contract events for offensive skill-position players, and the final model is estimated on the 369 observations with complete information for all regressors. The dependent variable is the log of real average annual value in 2024 dollars, which allows contracts signed in different years to be compared on a common purchasing-power basis. The explanatory variables include efficiency and usage, measured by expected points added per game and opportunities per game, position indicators, draft number, rookie-extension status, and the NFL salary cap in the year the contract was signed. By including position dummies and their interactions with efficiency, the model allows the market return to efficiency to differ across quarterbacks, running backs, wide receivers, and tight ends.

This is a worthwhile topic for at least two reasons. First, it sheds light on player evaluation and the market for talent in professional sports, which is often treated as a useful laboratory for studying labor-market behavior more broadly. Sports offer a rare setting in which

performance can be observed more directly than in many other industries, yet pay still depends on bargaining, institutional rules, and expectations about future output. Studying how NFL teams structure compensation can therefore offer broader insight into how firms reward workers. Second, the topic is practically relevant inside the football industry. Team owners and managers regularly make costly contract decisions, and an inaccurate assessment of a player's market value can tie up budget resources and weaken a team's competitiveness. A better understanding of the predictors of contract value can therefore improve both economic reasoning and football decision-making.

The econometric specification is designed to separate, as much as possible, different sources of a player's market value. Opportunities per game measure a player's centrality to his team's offensive strategy, while expected points added per game measure his contract-year productivity on a per-game basis. Draft number serves as a proxy for reputation and prior organizational investment, while rookie-extension status captures a distinct contractual setting negotiated outside the ordinary free-agent market. The salary cap proxies for league-wide budget capacity in the signing year. Using both position dummies and interactions between position and efficiency, the model allows the premium attached to efficiency to vary across quarterbacks, running backs, wide receivers, and tight ends.

The results indicate that contract value depends more strongly on usage, draft status, and contractual context than on a single uniform efficiency premium. In the full-sample specification, opportunities per game are positively and significantly associated with real average annual value, draft number is negatively associated with contract value, and signing a rookie extension predicts higher compensation. The interaction structure also suggests that the premium attached to efficiency is not constant across the four offensive skill positions. The overall explanatory power of the model is respectable, with an R-squared of approximately 0.515. Taken together, these patterns suggest that NFL teams pay for more than efficiency alone. They also pay for offensive centrality, prior investment, and the market setting in which the contract is negotiated. The economic framework, data, methodology, and results are discussed in detail below.

II. Theory/Framework

The economic theory underlying this paper begins with a simple idea: an NFL contract is the market price of expected future production. Teams do not pay for past performance as an end in itself. Instead, they use past performance, player characteristics, and league conditions to form expectations about what a player is likely to produce over the life of a contract. In that sense, the average annual value of a deal can be understood as a wage outcome in a specialized labor market. Because the NFL operates under a hard salary cap, with finite roster spots and relatively transparent performance measures, it provides a useful setting in which contract outcomes can be linked to objective indicators of productivity, usage, and scarcity.

Contract-year efficiency should matter because it captures how productive a player was when he was on the field. In this paper, expected points added per game is used as the main efficiency measure. EPA per game is useful because it goes beyond raw totals and measures contribution to offensive success. All else equal, a player who adds more expected points per game should appear more valuable to teams allocating scarce cap space. At the same time, the return to efficiency need not be identical across positions. Quarterbacks influence nearly every passing play and are typically the least replaceable offensive skill players, so a given increase in efficiency may command a different premium for quarterbacks than for running backs, wide receivers, or tight ends. For that reason, the model allows the relationship between EPA per game and contract value to vary by position through interaction terms.

Opportunities per game should matter as well because they capture how central a player is to an offense. In this paper, opportunities per game are defined as the average number of carries, pass attempts, and receptions per game in the contract year. In economic terms, this variable can be interpreted in two ways. First, usage may reflect how important a player is within the offense. Second, usage can proxy for the amount of information available to managers about the player's ability. Players with high-volume roles give teams more evidence that their performance is sustainable rather than driven by a small sample of favorable plays. Under those conditions, heavier usage may raise contract value because it signals responsibility, durability, and trust.

Draft number reflects a mechanism of reputation and status formation. Early draft picks enter the league with stronger prior beliefs about their talent, often receive more organizational

investment, and may retain status advantages even after recent performance is observed. For that reason, draft number should have a negative relationship with contract value. Rookie-extension status captures another mechanism. A player who signs a long-term extension before reaching unrestricted free agency is often a young player whom the team views as a long-run asset. That contractual setting may itself be associated with higher pay. Finally, the NFL salary cap in the signing year captures a market-wide force that can affect contract value. When the cap rises, teams collectively have more room to spend on labor, which should, in principle, raise the market price of talent.

Overall, five hypotheses follow from the model. First, the association between EPA per game and contract value should be positive, although its size may differ by position. Second, opportunities per game should be positively related to contract value. Third, draft number should affect contract value negatively. Fourth, rookie extensions should be associated with higher contract values. Fifth, the salary cap in the signing year should have a positive relationship with contract value. The regression model is constructed to evaluate these hypotheses jointly rather than in isolation.

III. Data Description

The data are collected by matching the financial details of 372 significant contract events for offensive skill-position players who signed market contracts between 2001 and 2024 with their contract-year performance and usage, along with a limited set of background and market variables. The unit of observation is the individual contract event rather than a season in a player's career, because the goal is to relate the terms of a specific contract to the player's performance and usage in the contract year. The sample is restricted to offensive skill-position players so that the comparison remains focused on positions that contribute to offensive production but differ sharply in role, replaceability, and market value. The resulting dataset includes 94 quarterbacks, 68 running backs, 130 wide receivers, and 80 tight ends.

The contract variables include information on average annual value from NFL contract records, while the performance variables are drawn from play-by-play data. In addition to nominal average annual value, the paper uses real average annual value adjusted to 2024 dollars.

Because contracts were signed across many different years, this inflation adjustment is necessary to make older and newer deals comparable. The gap between nominal and real values is substantial: the mean nominal average annual value in the sample is about \$11.75 million, while the mean real average annual value is about \$38.37 million in 2024 dollars. The regression uses the natural log of real average annual value as the dependent variable because the log transformation reduces the influence of extreme contracts and allows the estimates to be read in percentage terms.

Two main explanatory variables are used to capture different dimensions of player value. The first is EPA per game, which measures expected points added per game and ranges from -8.04 to 13.07, with a sample mean of 1.34. The second is opportunities per game, defined as carries, pass attempts, and receptions per game in the contract year; it ranges from 0 to 47, with a mean of 12.21. EPA per game captures how efficiently a player performs, while opportunities per game captures how heavily he is used. The model also includes interaction terms between position and EPA per game so that the market return to efficiency can differ across quarterbacks, running backs, wide receivers, and tight ends.

Several additional variables capture important aspects of contractual context. Draft number serves as an indicator of draft pedigree and ranges from 1 to 260, with a sample mean of 119.97. Rookie extension is a dummy variable equal to one when the contract was signed as a rookie extension and zero otherwise; there are 61 rookie extensions in the sample. Games played in the contract year indicate how much game information underlies the performance measures, with a mean of 13.15 games. Finally, the NFL salary cap in the signing year serves as a proxy for league-wide budget capacity. It ranges from \$67.4 million to \$255.4 million, with a mean of \$181.60 million. Because this variable is missing for three observations, the regression analysis is estimated on 369 observations.

IV. Data Analysis

The empirical analysis proceeds in two stages. The first stage is descriptive and graphical. Before estimating the regression model, the paper examines the distribution of contract values and the raw relationship between compensation, performance, usage, position,

and signing year. This step matters because the sample is not balanced across positions or contract sizes, and NFL pay data are visibly skewed by a relatively small number of very large deals. The histogram of real average annual value confirms that pattern: most contracts cluster in the lower and middle ranges, while a limited number of extreme observations—especially at quarterback—stretch the upper tail. That skewness is the main reason the regression uses the natural log of real average annual value rather than the level of the contract itself. The log transformation compresses extreme values and creates a more stable basis for comparing contracts that differ sharply in scale.

The remaining descriptive figures show preliminary evidence about the relationships later examined in the regression. The boxplot of real contract value by position shows that quarterbacks occupy a much higher compensation range than the other offensive skill positions. They have the highest median contract value and the widest dispersion. Running backs, tight ends, and wide receivers cluster at lower levels, although the data still suggest meaningful heterogeneity within those groups. The scatterplot of EPA per game against real average annual value points to a positive relationship between efficiency and pay, but the relationship is noisy and appears to vary by position. Many highly paid quarterbacks cluster at moderate efficiency levels, while players at other positions can post respectable efficiency without receiving quarterback-level contracts. That pattern supports the decision to interact EPA per game with position rather than assume a single market premium across all offensive roles.

The scatterplot of opportunities per game against real average annual value provides additional evidence about what drives NFL contract prices. It suggests a strong relationship between contract value and player usage, although that relationship differs by position. Because quarterback opportunities include pass attempts, quarterbacks appear in the upper usage range, while running backs, wide receivers, and tight ends occupy lower ranges. The figure suggests that there is a premium on being central to the offense. Players with heavier workloads tend to sign larger contracts, which is consistent with the idea that the market rewards more than pure efficiency. The line graph of mean real average annual value by position and year shows a similar pattern over time.

These descriptive results do not establish causation, but they clarify the patterns that motivate the regression analysis. The second stage of the paper is a multivariate regression in which the dependent variable is the log of real average annual value in 2024 dollars. The explanatory variables are contract-year EPA per game, opportunities per game, position dummies, EPA-per-game-by-position interactions, draft number, rookie-extension status, and the NFL salary cap in the signing year. This specification is intended to separate several components of contract valuation at the same time: whether teams reward efficiency after controlling for workload, whether workload matters after efficiency is taken into account, whether the premium to efficiency differs by position, whether draft order matters once recent performance is observed, whether rookie extensions affect pay, and whether league-wide budget capacity matters for contract values. Because the model is estimated only on observations with complete information, the regression compares players within a common and internally consistent sample.

The estimating equation for the main specification is:

$$\begin{aligned} \ln(\text{RealAAV}_i) = & \beta_0 + \beta_1 \text{EPA}_i + \beta_2 \text{Opps}_i + \beta_3 \text{Draft}_i + \beta_4 \text{RookieExt}_i + \beta_5 \text{Cap}_i \\ & + \gamma_1 \text{RB}_i + \gamma_2 \text{TE}_i + \gamma_3 \text{WR}_i \\ & + \delta_1 (\text{RB}_i \cdot \text{EPA}_i) + \delta_2 (\text{TE}_i \cdot \text{EPA}_i) + \delta_3 (\text{WR}_i \cdot \text{EPA}_i) + \varepsilon_i \end{aligned}$$

In this specification, i indexes contract events, the dependent variable is the log of real average annual value in 2024 dollars, quarterback is the omitted position category, and the interaction terms allow the association between efficiency and contract value to vary by position. Equation (1) is estimated by ordinary least squares on the 369 observations with complete information for all included variables, so the reported coefficients are interpreted as conditional percentage associations rather than causal effects.

This empirical framework is not intended to identify causal effects. Contract negotiations depend on many factors that are not directly observed in the data, including injury risk, age trajectories, scheme fit, locker-room value, and private team evaluations. Even so, the regression is a useful tool for identifying the observable factors most strongly associated with the real value of NFL offensive skill-position contracts. In that sense, the model should be interpreted as a conditional-association framework grounded in economic reasoning.

V. Main Results

Table 2 in Appendix 1 presents the main OLS results for the full-sample specification. The regression is estimated on 369 contract events with complete information for all variables in the model. Because the dependent variable is the log of real average annual value in 2024 dollars, the coefficients can be interpreted as conditional percentage associations, holding the other included variables constant. Overall, the model fits the data reasonably well for a cross-sectional study of professional sports contracts. The F-statistic is 34.45, the p-value is effectively zero, the R-squared is 0.515, and the adjusted R-squared is 0.500. In practical terms, that means the included variables explain a substantial share of the variation in contract value, even though a meaningful portion of contract negotiations still depends on factors that are not directly observed in the dataset.

One of the key results is that efficiency matters, but not in a uniform way across positions. The baseline coefficient on contract-year EPA per game is positive at 0.0588 and statistically significant at the 1 percent level ($p = 0.0098$). Because quarterbacks are the omitted category, this coefficient captures the conditional association between higher contract-year efficiency and higher real contract value for quarterbacks. Interpreted approximately, a one-unit increase in EPA per game is associated with about a 5.9 percent increase in contract value for quarterbacks. That is an economically meaningful premium, especially given the central role quarterbacks play in offensive production and roster construction.

The interaction terms show that the return to efficiency is not the same across the entire market for offensive skill players. The interaction between wide receiver status and EPA per game is positive and statistically significant (0.1044, $p = 0.0297$). Combined with the quarterback baseline, this implies a slope of roughly 0.163 for wide receivers. By contrast, the interaction terms for running backs and tight ends are not statistically significant, which suggests that their efficiency premiums are not obviously distinguishable from the quarterback baseline within this sample. Substantively, the estimates indicate that the market places an especially strong premium on wide receiver efficiency, while the pricing of running back and tight end efficiency is less precise once the other controls are included.

The strongest result in Table 2 is the coefficient on opportunities per game. Its estimated value is 0.0608 and it is highly statistically significant ($p < 0.001$). In approximate percentage terms, each additional opportunity per game is associated with about a 6.1 percent increase in real contract value, holding constant efficiency and the other variables in the model. That is a substantial premium, which implies that teams are paying not only for efficiency but also for offensive centrality. In other words, they are paying not just for how effective a player is, but also for how central he is to the offense.

Draft status also remains economically important after controlling for recent performance and usage. The coefficient on draft number is negative at -0.00266 and highly statistically significant ($p < 0.001$). Because higher draft numbers correspond to later selections, the negative sign implies that earlier draft picks tend to sign larger contracts. The one-pick effect is modest, but it accumulates across meaningful draft ranges. This result suggests that even after performance is taken into account, signaling and reputation continue to matter in the labor market for NFL offensive skill players.

The coefficient on rookie-extension status is also positive and highly statistically significant at 0.6318 ($p < 0.001$). In log terms, that corresponds to about a 63 percent difference in contract value; using the exponential transformation, the implied premium is roughly 88 percent. It is one of the largest coefficients in the model. This supports the view that rookie extensions occur in a distinct bargaining environment. Teams appear willing to pay substantial premiums to lock in players they view as long-run offensive cornerstones.

Somewhat unexpectedly, the coefficient on the NFL salary cap in the signing year is negative and statistically significant (-0.00830, $p < 0.001$). This runs against the simple intuition that a larger cap should lead to higher player pay. However, once the dependent variable is converted into real 2024 dollars, the coefficient may also absorb time-related structure in the sample rather than a pure salary-cap effect. For that reason, the negative sign is more plausibly interpreted as a conditional time-related association than as a causal estimate of budget expansion on contract prices.

Position categories also matter even after accounting for performance and usage. Relative to quarterbacks, the coefficients on running back, tight end, and wide receiver are all positive and statistically significant. Because the model includes interaction terms, these coefficients should be interpreted as position differences when contract-year EPA per game equals zero. Even with that caveat, they reinforce the broader conclusion that the market treats the four positions differently. Position is not simply a label; it changes how the market translates production into compensation.

The model's predictive performance also appears to differ across positions. The relatively strong and statistically significant coefficients on efficiency and interaction effects for quarterbacks and wide receivers in Table 2 suggest that the model may capture some determinants of contract value more definitively for these positions. In particular, the baseline EPA coefficient for quarterbacks (0.0588, $p = 0.0098$) and the additional premium for wide receivers through the interaction term (0.1044, $p = 0.0297$) indicate that both efficiency and its positional value are estimated more precisely for these groups, which may contribute to somewhat more reliable predictions within this specification. By contrast, the interaction terms for running backs (-0.0336, $p = 0.6646$) and tight ends (0.0322, $p = 0.7378$) are not statistically significant, suggesting that the model does not estimate the return to efficiency for these positions as clearly. As a result, predictions for running backs and tight ends may deviate more from observed values because their contract valuation may depend more heavily on factors not included in the model, such as injury risk and career longevity for running backs, blocking responsibilities and scheme fit for tight ends, and situational usage that is not fully captured by opportunities per game.

To further evaluate overall model fit, Figure 7 compares predicted and actual log real contract values. The model demonstrates a reasonably strong fit, consistent with the reported R-squared of 0.515, as many observations cluster around the 45-degree line. However, the dispersion around that line is nontrivial, indicating that prediction errors remain economically meaningful. In particular, several higher-value contracts are overpredicted, while some lower-value contracts are underpredicted, suggesting that the model does not fully capture the upper and lower tails of the contract distribution. These discrepancies point to omitted factors that are important in contract negotiations but not included in the regression, such as player age at

signing, prior injury history, contract length and guarantees, team salary cap constraints beyond the aggregate cap level, and market-specific dynamics such as positional scarcity or free agency competition. Figure 8 provides additional context by comparing mean EPA per game in the contract year to post-contract performance across positions. The results show that, for most positions, post-contract EPA is slightly higher than contract-year EPA, particularly for wide receivers and tight ends, while running backs exhibit a small decline. This pattern suggests that teams may be successfully identifying players whose performance is expected to remain stable or improve, although the decline for running backs is consistent with concerns about positional durability and shorter career peaks.

Taken together, the results provide support for the paper's central argument. NFL teams do not appear to pay for efficiency alone. They pay for a broader bundle of traits that includes efficiency, offensive centrality, draft pedigree, and contractual context, and the importance of those traits differs across positions. The most robust findings are the positive association between opportunities per game and contract value, the negative association between later draft position and contract value, and the positive association between rookie-extension status and contract value.

VI. Conclusions

This paper examined the observable determinants of the inflation-adjusted annual value of NFL contracts for offensive skill-position players. Using a cross-section of 372 contract events and estimating the final model on the 369 observations with complete data, the analysis related contract value to contract-year opportunities, contract-year efficiency measured by EPA per game, position, draft number, rookie-extension status, and the NFL salary cap in the signing year. Overall, the results show that NFL contract valuation cannot be reduced to a single performance statistic. Compensation instead reflects a broader set of characteristics, including efficiency, offensive centrality, prior status, and contractual context.

Several findings stand out. Opportunities per game are one of the strongest predictors of contract value, which suggests that players who occupy central offensive roles receive substantial market premiums. Draft number remains negatively associated with contract value

even after controlling for recent performance and usage, which implies that draft status still matters in the labor market for NFL talent. Rookie extensions are associated with much larger contract values, consistent with the idea that teams use those contracts to secure players they view as long-run cornerstones. Efficiency also matters, but its marginal return differs across positions. The return to contract-year EPA per game appears strongest for quarterbacks and wide receivers, while the estimates for running backs and tight ends are less precise once the full set of controls is included.

More broadly, the paper suggests that even in a high-information labor market, compensation reflects more than measured productivity alone. NFL teams appear to price offensive skill-position contracts using efficiency, usage, draft status, and institutional context together rather than any single metric in isolation. At the same time, the results should be interpreted cautiously because the paper identifies conditional associations rather than causal effects. Contract negotiations depend on many factors that are not fully observed in the dataset, including age, health expectations, leadership value, scheme fit, and internal team evaluations.

The salary-cap coefficient likely captures some time-related structure after all contracts are converted into real 2024 dollars. Future research could extend the analysis by including variables related to age, injuries, experience, guaranteed money, team characteristics, and contract type. Even with those limitations, the paper provides a solid answer to the core question: NFL teams appear to price offensive skill-position contracts using a combination of efficiency, usage, status, and institutional considerations.

VII. References

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Appendix 1. Figures and Tables

Table 1. Summary Statistics

Variable	N	Mean	SD	Min	Max
Nominal AAV (Millions)	372	11.749	11.946	0.750	60.000
Real AAV (Millions, 2024 Dollars)	372	38.367	45.725	0.828	239.602
Log Real AAV (2024 Dollars)	372	2.948	1.306	-0.189	5.479
Draft Number	372	119.970	98.364	1.000	260.000
Games Played, Contract Year	372	13.153	4.977	1.000	20.000
EPA per Game, Contract Year	372	1.340	2.773	-8.038	13.066
Opportunities per Game, Contract Year	372	12.214	13.839	0.000	47.000
NFL Salary Cap, Signing Year	369	181.604	43.037	67.400	255.400

Note: The sample contains 372 contract events for NFL offensive skill-position players. NFL salary cap information is available for 369 observations. Contract values are reported in millions of U.S. dollars, with real values adjusted to 2024 dollars.

Table 2. OLS Regression Results: Determinants of Log Real AAV

Variable	Coefficient	Std. Error	t-Statistic	p-Value	Lower 95%	Upper 95%
EPA per Game (QB baseline)	0.0588	0.0226	2.5978	0.0098	0.0143	0.1033
RB	0.4400	0.2074	2.1210	0.0346	0.0320	0.8480
TE	1.0507	0.2624	4.0038	0.0001	0.5346	1.5668
WR	0.9056	0.2503	3.6188	0.0003	0.4135	1.3978
RB x EPA per Game	-0.0336	0.0773	-0.4339	0.6646	-0.1857	0.1185
TE x EPA per Game	0.0322	0.0962	0.3350	0.7378	-0.1569	0.2214
WR x EPA per Game	0.1044	0.0478	2.1831	0.0297	0.0104	0.1985
Opportunities per Game	0.0608	0.0075	8.0964	0.0000	0.0460	0.0756
Draft Number	-0.0027	0.0006	-4.7687	0.0000	-0.0038	-0.0016
Rookie Extension	0.6318	0.1422	4.4423	0.0000	0.3521	0.9115
NFL Salary Cap, Signing Year	-0.0083	0.0012	-7.1304	0.0000	-0.0106	-0.0060
Constant	3.1545	0.3370	9.3597	0.0000	2.4917	3.8173

Note: Dependent variable is log real annual average value in 2024 dollars. Omitted position category is quarterback. The regression is estimated on 369 observations with complete information for all included variables.

Table 3. Model Fit Statistics

Statistic	Value
Number of observations	369
F statistic	34.4519
Prob > F	0.0000
R-squared	0.5149
Adjusted R-squared	0.5000
Root MSE	0.9224
Residual degrees of freedom	357

Figure 1. Distribution of Inflation-Adjusted Contract AAV

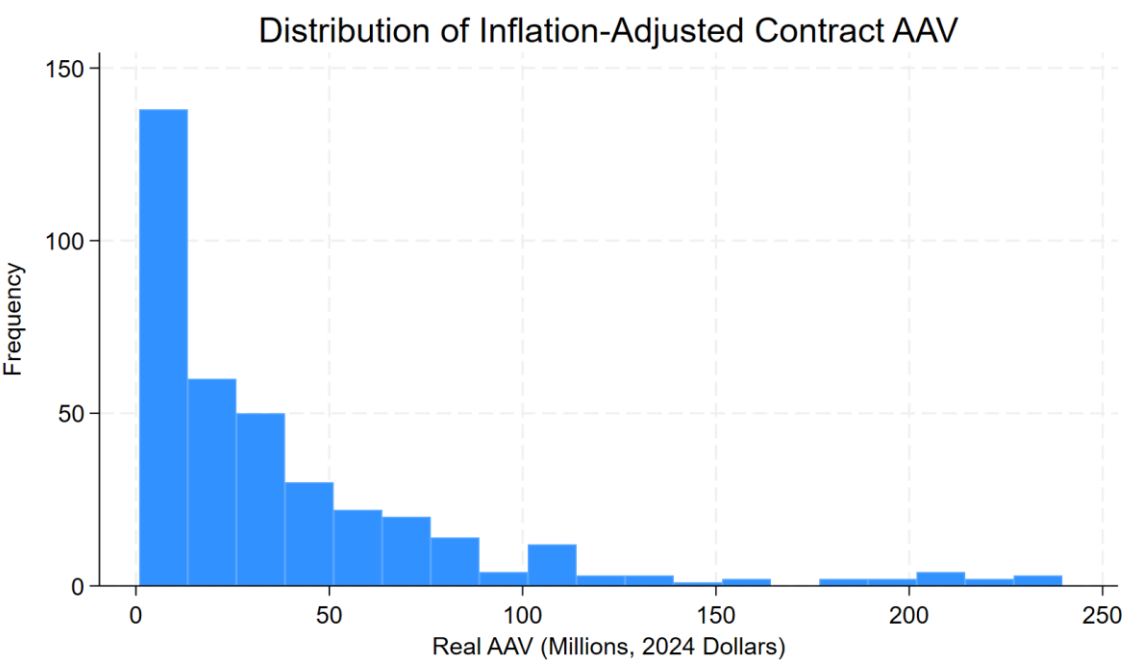


Figure 2. Distribution of Log Real AAV

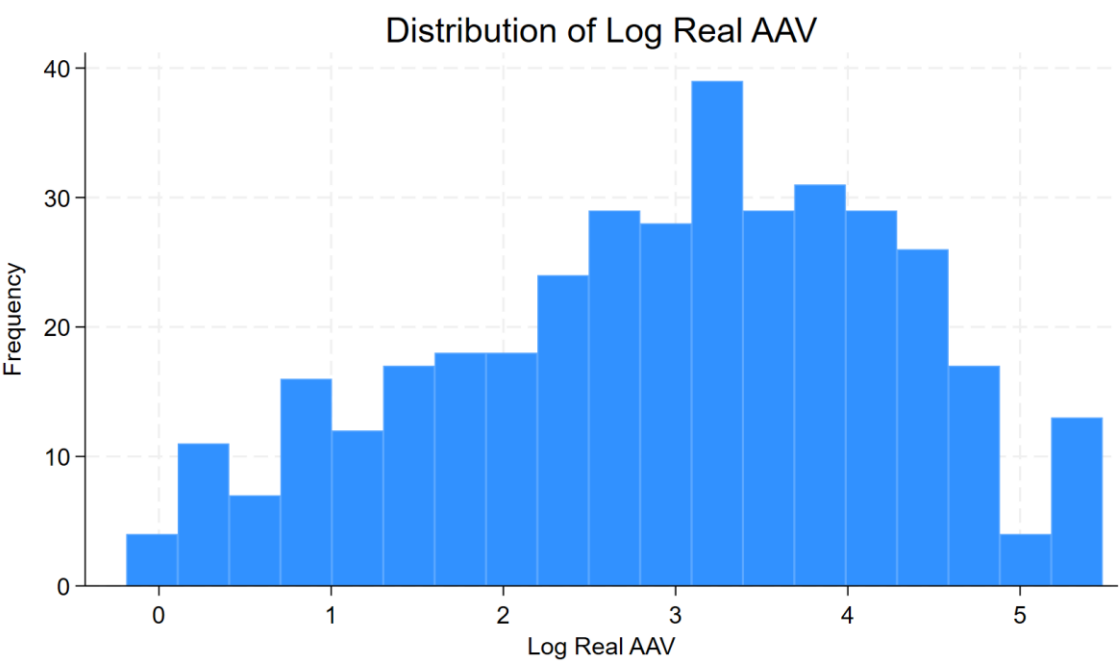


Figure 3. Real AAV by Position

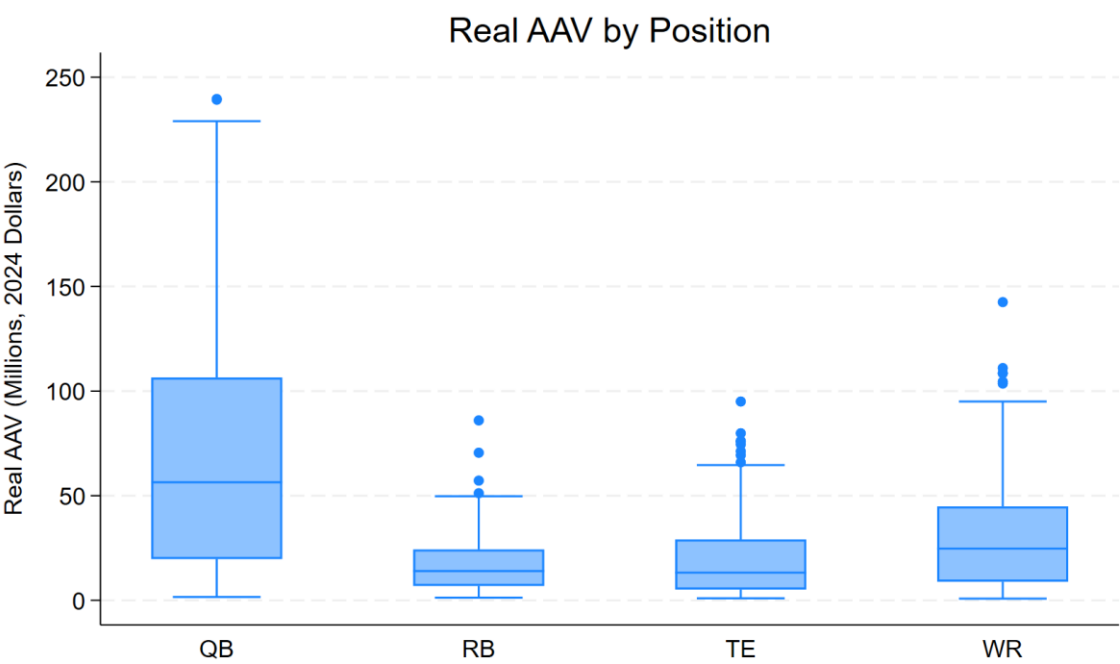


Figure 4. Contract-Year EPA vs. Real AAV

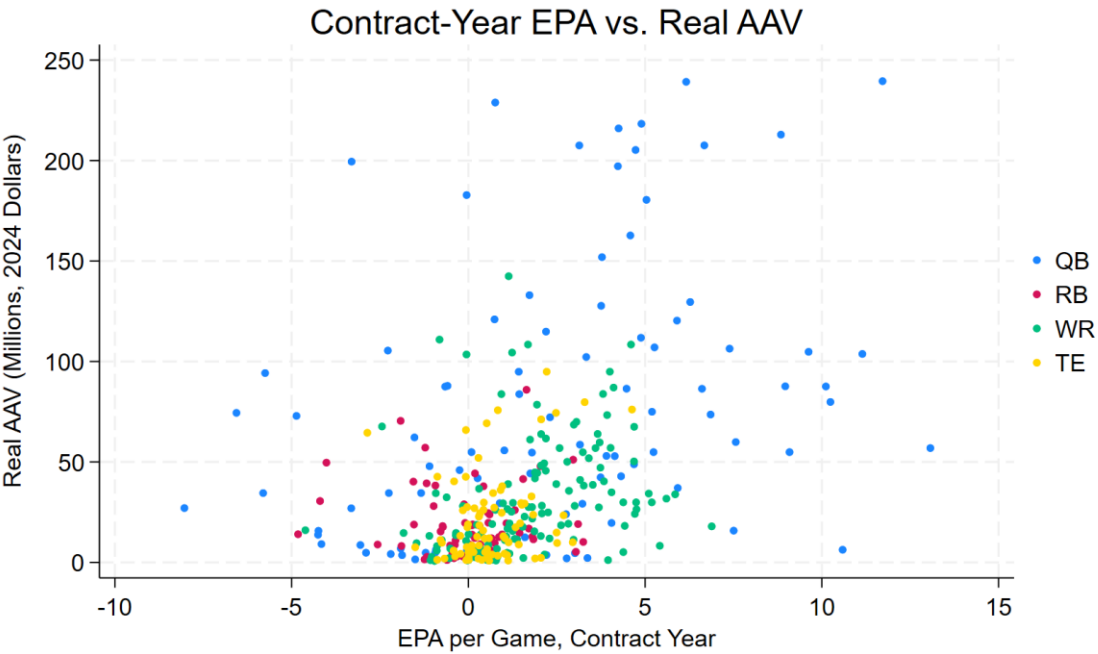


Figure 5. Contract-Year Opportunities per Game vs. Real AAV

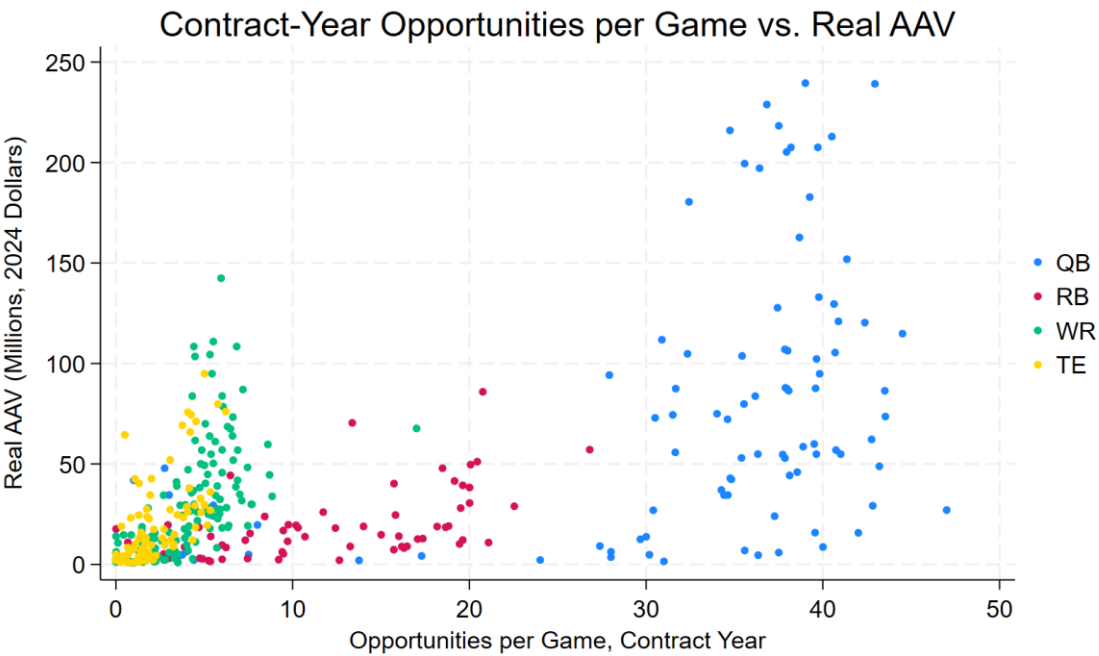


Figure 6. Mean Real AAV by Position Over Time

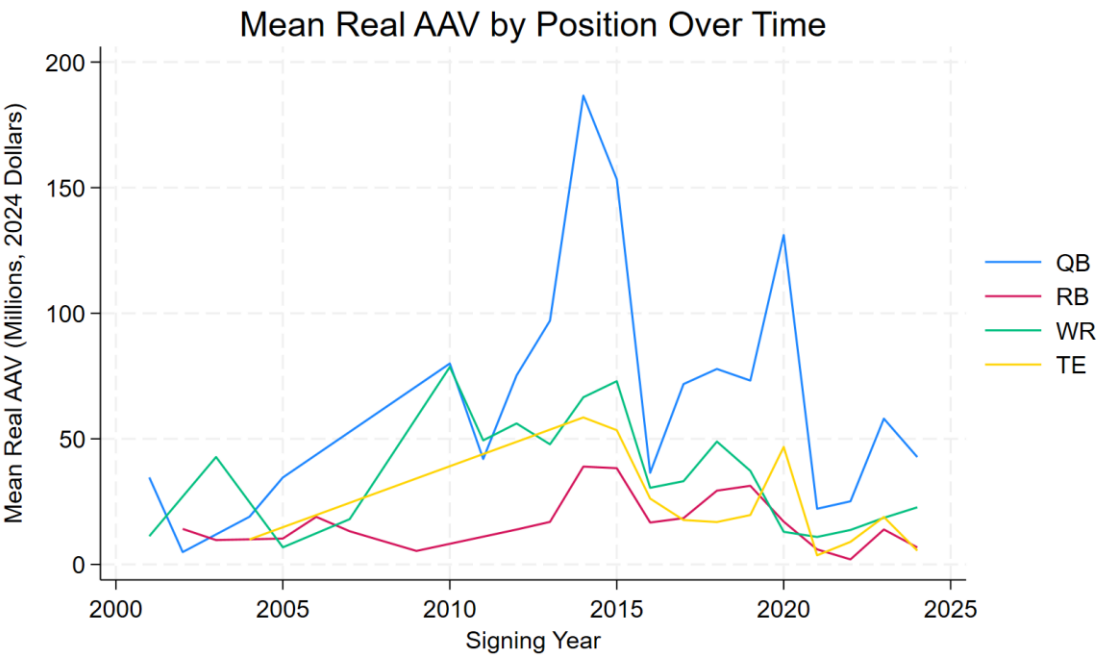


Figure 7. Predicted vs. Actual Log Real AAV

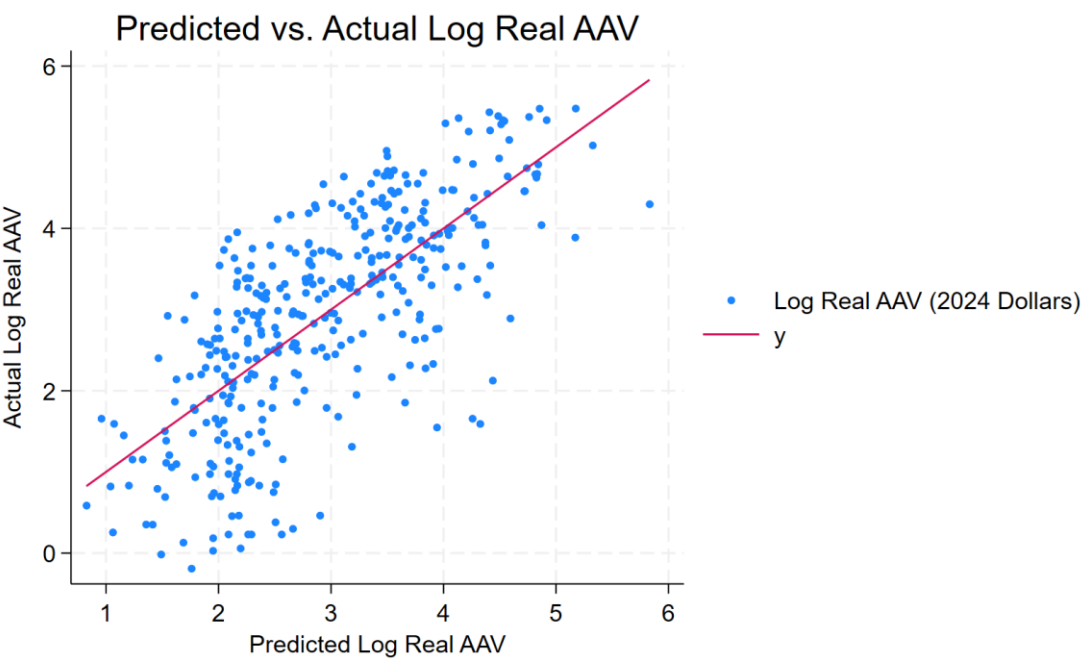


Figure 8. Mean EPA per Game in Contract Year vs. Post-Contract by Position

